

Chapter 22

Applications Of Inversive Geometry

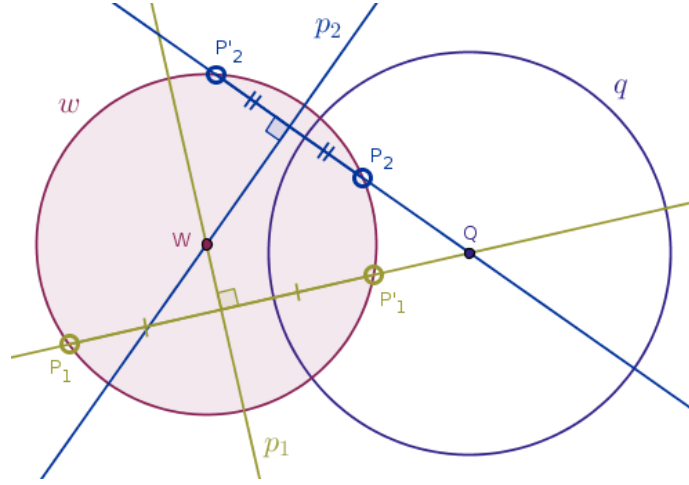
Intersecting A Circle Under An Angle With Constraints

Now that we have studied the mechanics of solving problems in reverse order or in reverse time line in the Verbal College Entrance Exams Chapter, 17, let us solidify our understanding of this particular problem-solving approach by exercising it on a problem of constructing a circle that intersects a given circle under a given angle *with constraints*. More specifically, let us solve the following problem, with, traditionally, Euclidean tools, a compass and a straight edge, alone.

Problem 28. Construct a circle that passes through two given distinct fixed points, A and B , and intersects a given (coplanar) circle, s_0 , under a given angle, θ_0 .

Solution. Recall that in the Problem 15 from Chapter 11 we were asked to construct a circle that is *orthogonal* to a given circle and that passes through two given fixed distinct points.

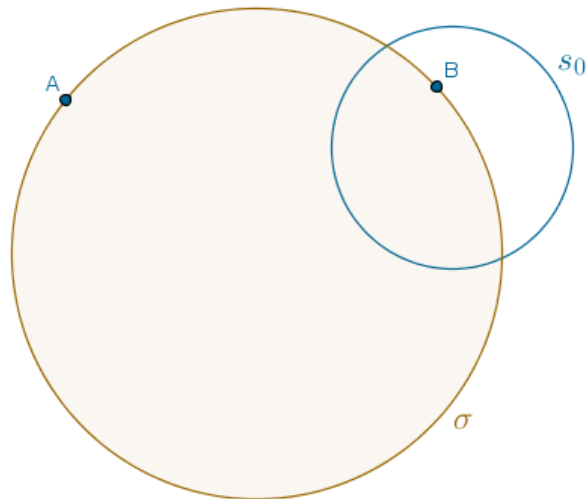
In our solution of that problem we named the sought-after circle as w , the given circle was named q and the given and fixed points were P_1 and P_2 (Figure 11.7):



Thus, we have already dealt with an objective that is quite similar to the current one with the exception that the given angle θ_0 was fixed at $\pi/2$ radians. It follows, then, that in this case we would like to find a generic solution to the given problem when the given angle θ_0 varies as follows:

$$0 < \theta_0 < \frac{\pi}{2}$$

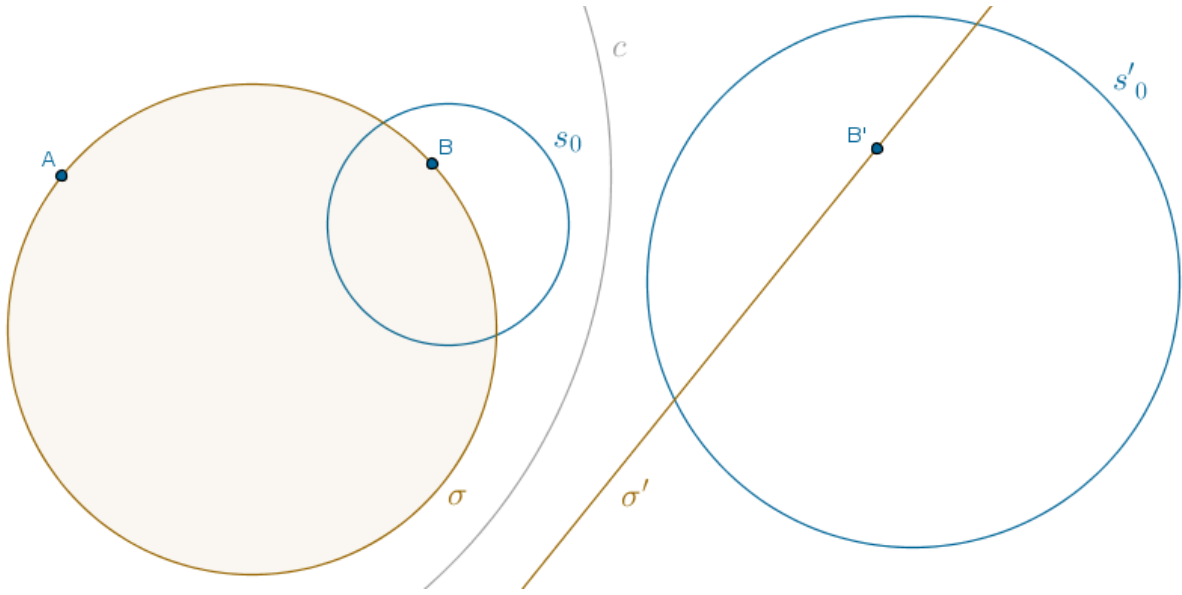
if we agree that for the magnitude of the angle θ_0 between any two intersecting straight lines we always choose the smallest one. This problem, thus, becomes a typical candidate for the application of the reverse-order problem-solving approach. Hence, we, first, assume that the mystery circle, σ , has been constructed somehow (Figure 22.1):



where, for the record, we have used $\theta_0 = 5\pi/12$ radians, for demonstration purposes.

Consider the inversion of our plane that houses all the input objects with respect to the circle $c(A, r)$ centered at the point A with a non zero radius r with positive power.

Under the said inversion, the circle s_0 will be taken into a circle s'_0 . The point B will be taken into its image point B' and, that is the key, the mystery circle σ will be taken into a straight line σ' that passes through the point B' (Figure 22.2):



Still going with the reverse-order flow of causality, using the objects s'_0 , σ' and B' , we would like to unwind the current configuration into a state, starting from which, this problem can be solved in the sought-after, forward flow of causality, order.

Tactically, what we are about to achieve can be referred to as a solution of *an inverted or a simpler problem*, as suggested by the idea explained in the Problems 01 Chapter, 10.

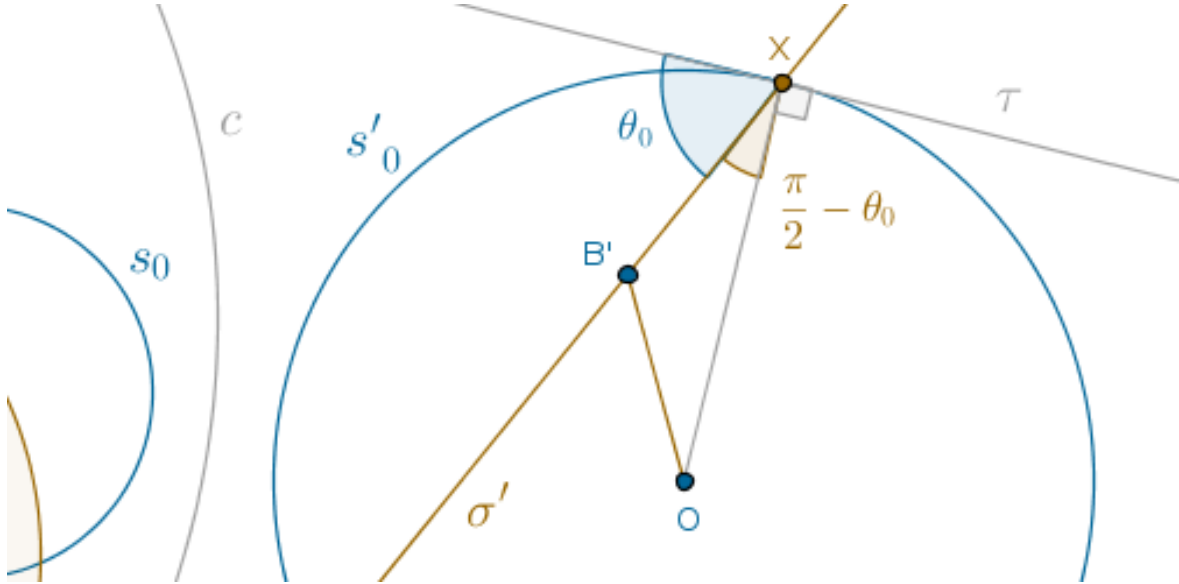
Put differently but equivalent, we now want to solve the following problem: intersect a given circle, s'_0 , by a straight line, σ' , passing through a given point, B' , under a given angle, θ_0 . And this, we reckon, is where we gain a competitive advantage because that problem is *simpler* than the original.

If we have a circle, such as the circle s'_0 , then the task of locating its center is *primitive*. Primitive in the sense that it is a previously-solved problem, left to the reader as an exercise.

Once the center, O , of the circle s'_0 is known, we can unpack the definition of the angle between two cooperating curves.

In this case, the angle θ_0 between the straight line σ' and the circle s'_0 is the (smallest) angle between the said straight line σ' and the tangent straight line, τ , drawn to the circle s'_0 through a point where these two geometric objects intersect.

If we agree to name one such point as X then we shall have (Figure 22.3):



Why do we insist on the fact that the angle between the reflected objects, σ' and s'_0 , will be θ_0 ? Because, according to the theorems 32–36, the magnitude of the angle θ_0 , but not its orientation, under the said inversion will be preserved. But that solves our problem completely. Why.

If the angle between the tangent τ and the straight line σ' is θ_0 then the angle $\angle OXB'$ under which the line segment OB' is seen from the mystery point X will be fixed and constant, $\pi/2 - \theta_0$.

However, the exact location of the point X on the circle s'_0 can be recovered via the method of intersecting loci. Such a problem is, again, primitive or a previously solved one.

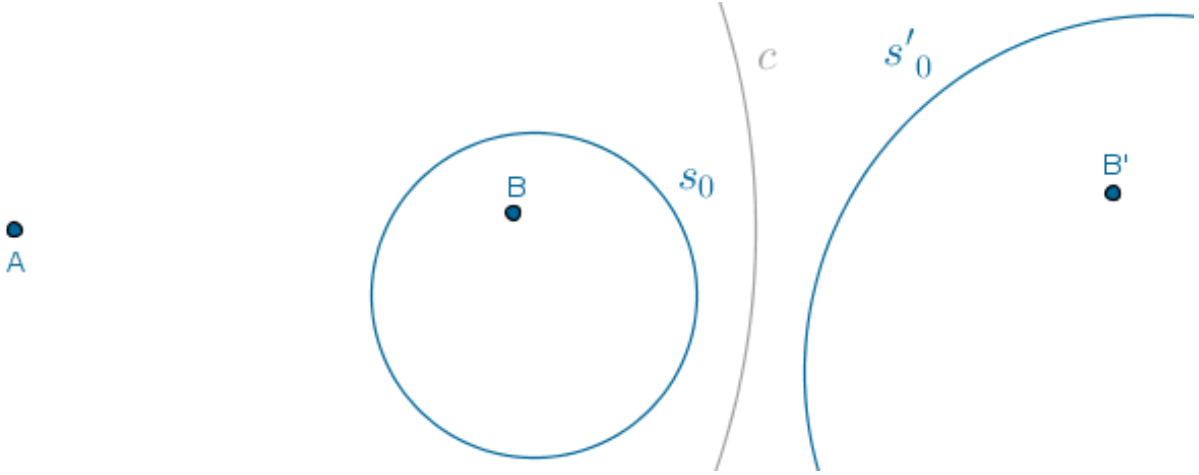
Lastly, if we now remove the mystery circle σ and the mystery straight line σ' from our reverse-order reasoning then we will be left with:

- the known point B'
- the known circle s'_0
- the known angle $\pi/2 - \theta_0$

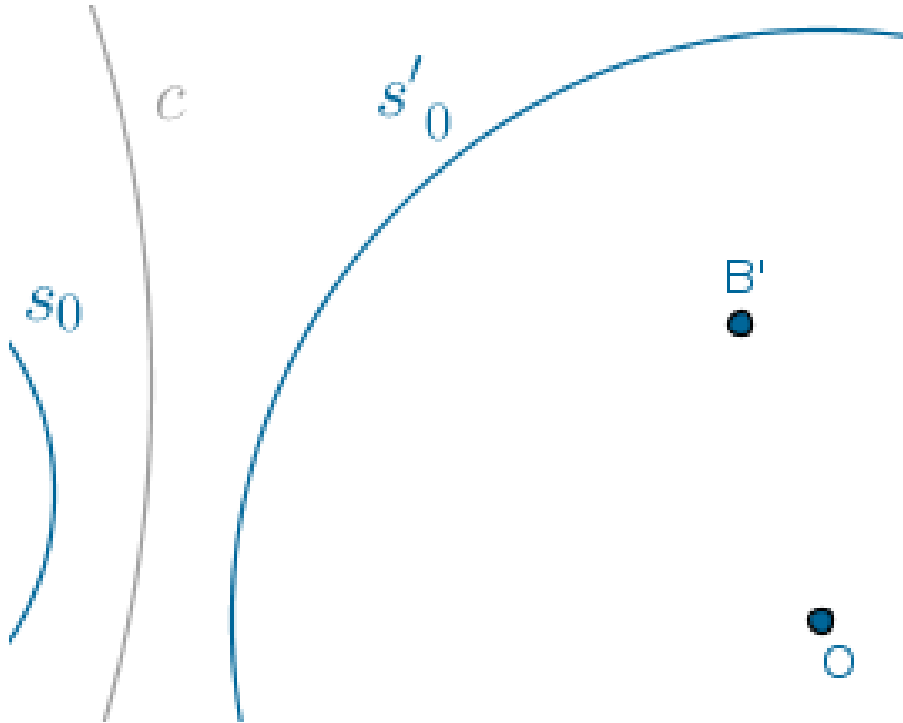
and the fact that the exact location of the point X on the known circle s'_0 is recoverable (with Euclidean tools alone). Thus, intuitively, the inverted objects listed above represent the original problem statement because we can always reflect these objects into their original images via a transformation described earlier.

That is why the state of the problem shown in Figure 22.3 is terminal or final, since we can now reverse the flow of causality to its opposite and generate a sought-after forward-order solution.

Step 1. Reflect the given point B and the given circle s_0 in the circle $c(A, r)$ with positive power into their images B' and s'_0 (Figure 22.4):

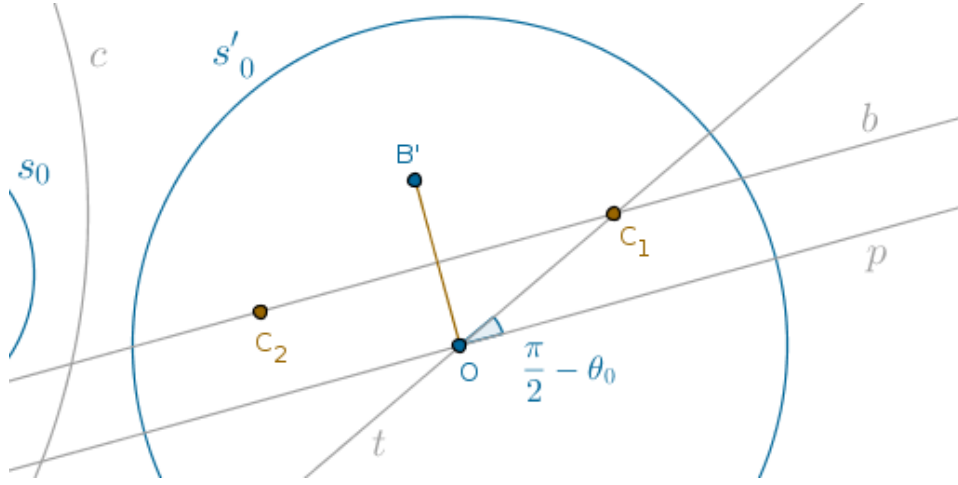


Step 2. Locate the center, O , of the reflected circle, s'_0 (Figure 22.5):

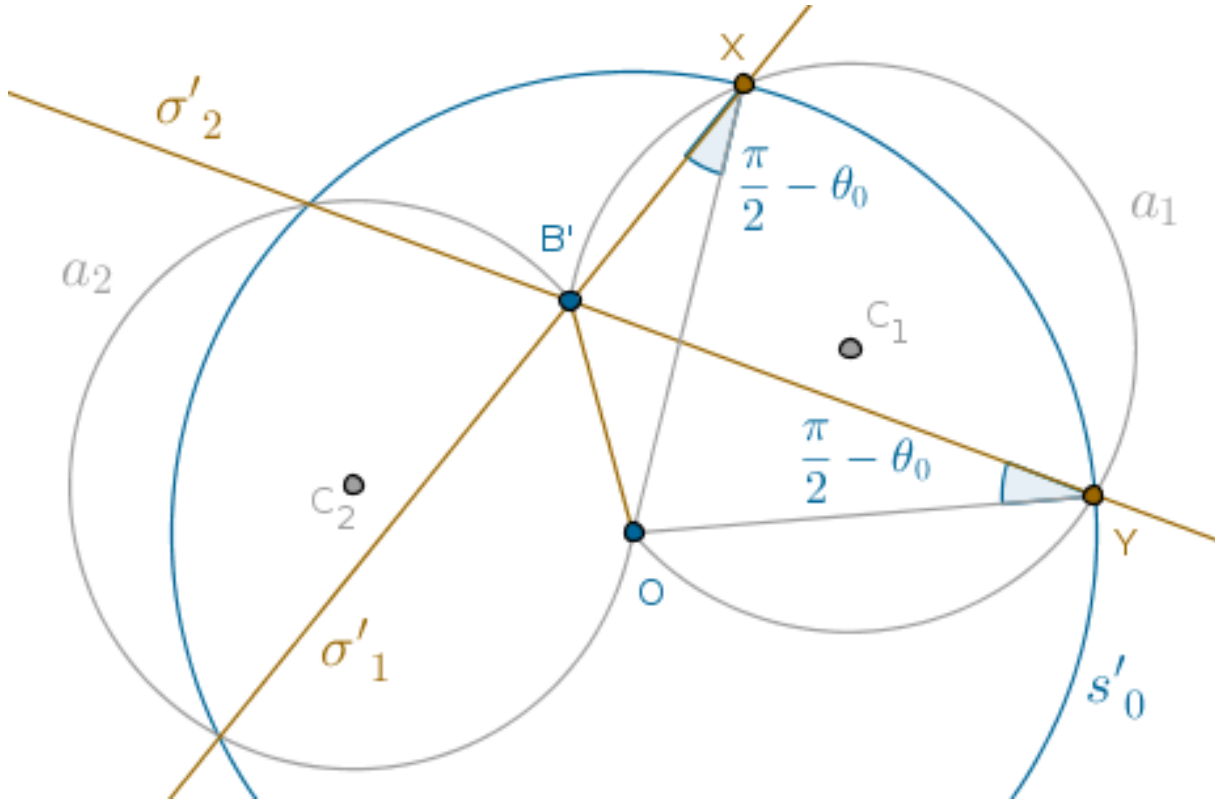


Step 3. Bisect the line segment OB' with a straight line, b ; draw a straight line, p , parallel to b through the point O ; construct the given angle $\pi/2 - \theta_0$ at the vertex O via a straight line, t , until that straight

line intersects the straight line b at the point C_1 ; reflect the point C_1 in the straight line OB' into the point C_2 (Figure 22.6):



Step 4. Draw one of two constant-angle circular arcs, a_1 or a_2 . The arc a_1 is centered at the point C_1 and has the radius $|C_1O|$ and the arc a_2 is centered at the point C_2 and has the radius $|C_2O|$ (Figure 22.7):



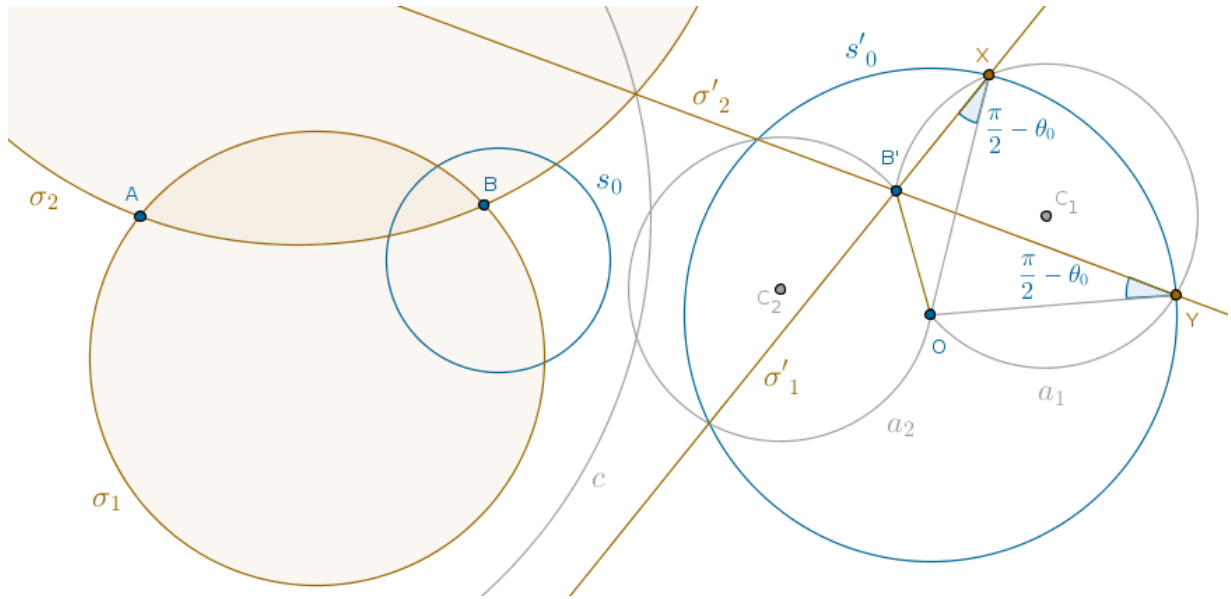
Since both constant-angle circular arcs $a_{1,2}$ will produce the same set of solutions, we will consider only one such arc, namely, the arc a_1 .

If the constant-angle circular arc a_1 has exactly two points, X and Y , in common with the reflected circle, s'_0 , as shown in the figure above, then these two points will define two distinct straight lines, σ'_1 and σ'_2 , that pass through the point B' and intersect the reflected circle, s'_0 , under the given angle θ_0 and the problem, thus, has two distinct solutions.

If the constant-angle circular arc a_1 has exactly one point in common with the reflected circle, s'_0 , then such a point will define a single straight line σ' that passes through the point B' and intersects the reflected circle, s'_0 , under the given angle θ_0 and the problem, thus, has only one solution.

If the constant-angle circular arc a_1 has exactly zero points in common with the reflected circle, s'_0 , then the problem has no solutions. As such.

Step 5. Reflect the straight lines σ'_1 and σ'_2 or just the straight line σ' , as the case may be, with respect to our circle $c(A, r)$ with positive power into the corresponding solution(s) sought-after, the circles σ_1 and σ_2 or just one circle σ (Figure 22.8):



In the figure above the solution circles σ_1 and σ_2 pass through the points A and B and intersect the circle s_0 under the angle θ_0 .

A Short Analysis

1. Observe how the solutions of *primitive* problems played the roles of the building blocks of a solution of a more complex problem. Primitive problems or primitives objects exist elsewhere in mathematics. Primitive Pythagorean Triples, for example, are the building blocks of an infinitude of generated by

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them Pythagorean Triples. In the theory of pencils of circles there exist *primitive elliptic*, *hyperbolic* and *parabolic pencils* of circles and so on.

2. Note how our “transform-solve-reverse transform” idea explained in Chapter 10 worked in our favor, again.